Benchmarking Multi-View Tree-Level Oil Palm Bunch Counting Under a Fixed Detector

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***Abstract*—In oil palm plantations, each harvest cycle is planned from a per-tree count of fruit bunches grouped by maturity class, since each class ripens on a different schedule. Multi-view imaging recovers bunches hidden by occlusion, but the same bunch then reappears across several side views, so per-image detections must be aggregated into a per-tree count rather than summed. Comparing methods for this tree-level aggregation remains underexplored in oil palm. This paper evaluates five regression counters and eight feature configurations on 953 oil palm trees and 3,992 side-view images, under two detection conditions: ground-truth detections, which measure the counters in isolation, and a fixed YOLO26m detector, which reflects a deployed pipeline. With ground-truth detections, the best reported counter reaches 98.05% Class ±1 accuracy and 92.20% Tree ±1 accuracy; under the fixed detector both fall sharply, to 77.48% and 32.62%. A tree-level bootstrap places the class-level gap at 20.57 points (95% CI 17.20–23.94), while no two counters differ significantly under fixed inputs on Class ±1 accuracy after multiplicity correction. Diagnostic matching finds 22.7% of test bunches undetected in every view, alongside substantial maturity-class confusion; these diagnostics do not provide an additive decomposition of the deployed error. The GT advantage persists in matched cross-validation, four YOLO26 checkpoints, and a separately trained YOLO11m, whose matched-counter GT gap is 23.94 points. These results support prioritising detector output quality on this dataset, while leaving improvements from other counters and detector families open.**

Keywords*—*Oil palm, fresh fruit bunch, black bunch census, multi-view counting, object detection, YOLO, tree-level regression, agricultural computer vision.

I. Introduction

Harvest planning in commercial oil palm plantations depends on knowing, for each surveyed tree, how many fruit bunches belong to each operational maturity class. The Black Bunch Census (BBC) formalises this requirement into a four-class count vector, B1 through B4, where each class carries distinct timing and logistics implications [1]. Automating BBC through image-based counting would reduce the cost and subjectivity of manual field surveys. An automated census is therefore useful only if it recovers the full per-class breakdown for each tree.

Tree-level counting is geometrically harder than image-level detection [2]. Bunches surround the full circumference of an oil palm, so a single image misses bunches due to occlusion, viewing angle, or overlapping fronds. Multi-view acquisition reduces missed bunches by photographing each tree from four to eight side views, but the same physical bunch then appears in several images. Fig. 1 illustrates the effect: a single B3 bunch is visible in three adjacent side views of the same tree. Naive summation of per-image detections counts appearances rather than bunch identities and is biased upward on this dataset.

Multi-view deduplication has been studied across several crops. Mango systems associated detections across sequential passes using navigation data and canopy geometry [3], and apple pipelines built row-level maps from overlapping two-side scans [4]; a comparative evaluation of the latter reports that counting errors originate mainly in detection quality rather than in the aggregation step [5]. Sparse and calibrated viewpoints have been addressed through correspondence-free triangulation [6] and few-view 3D reconstruction [7], and neural-field methods count from unstructured photographs in a shared 3D representation [8]. Most of these systems depend on dense sequential overlap, a moving sensor platform, or calibrated geometry. Their acquisition settings differ from the sparse tree-level views evaluated here.

Regression-based aggregation instead maps detection statistics to a tree-level count vector, an approach well established in agricultural vision [2], [4], [5], with YOLO-style detectors as the standard front-end [9], [10]. A counter is typically evaluated only on the detector output available during the study, so its reported accuracy reflects that detector's limitations as much as its own quality. Benchmarking counting methods on their own terms therefore calls for a detection-controlled condition alongside the realistic one.

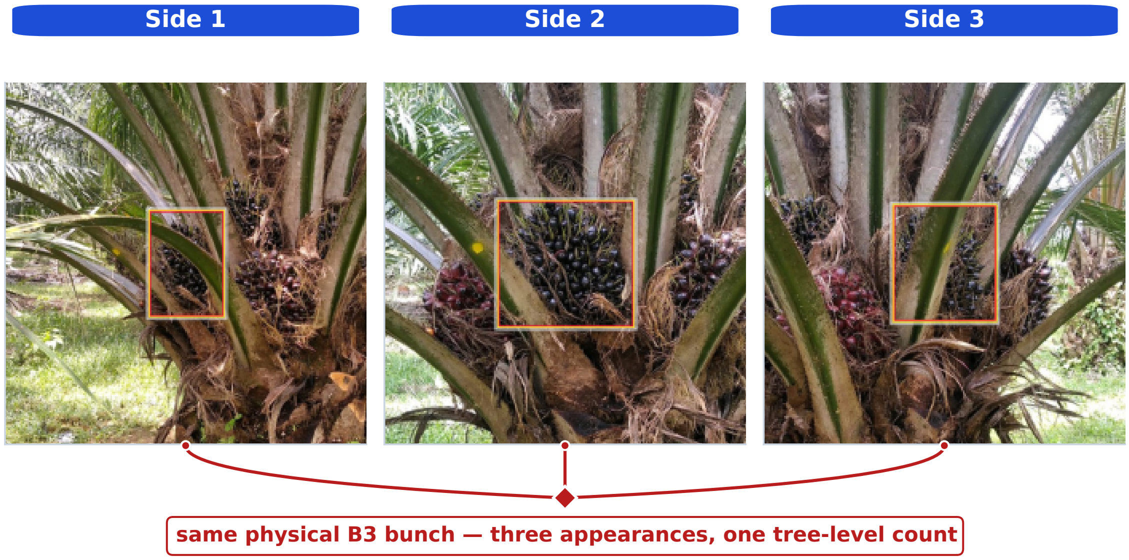

Fig. 1. A single physical B3 bunch is visible across three adjacent side views. If every per-image detection is counted, three appearances are added for one bunch. Tree-level counting must therefore aggregate evidence across views rather than sum it.

Oil palm counting has received less systematic study than apple or mango. Detectors for fresh fruit bunches (FFB) and maturity classification reach high image-level accuracy on ripe bunches but show recurring weaknesses on partially occluded and maturity-ambiguous classes [11], with both two-stage and YOLO-style architectures explored [12]–[15]. The closest public oil palm resource is a video dataset annotated at the frame level for detection [16]; it carries no tree-level counting labels and no counting evaluation has been reported on it. No existing benchmark evaluates tree-level counting strategies for oil palm under both controlled and realistic detection conditions.

This paper closes that gap by benchmarking tree-level multi-view BBC counting under two detection conditions on the SawitMVC dataset (953 trees, 3,992 images) [17]: ground-truth detections, which isolate counter performance from detector error, and a fixed YOLO26m detector, which reflects a deployed pipeline. Five regression models and eight feature configurations are compared, with feature ablation focused on fixed-detector inputs. The contributions are (1) a controlled-input reference and deployment baseline with tree-level confidence intervals and paired tests; (2) a feature ablation quantifying the limited gains observed among the tested representations; (3) appearance-level error analysis and diagnostic counting rules using ground-truth identities; and (4) sensitivity analyses across matched cross-validation, estates, YOLO26 checkpoints, and confidence thresholds.

II. Methodology

A. Dataset and Task Definition

The benchmark uses the SawitMVC multi-view dataset: 953 oil palm trees, 3,992 side-view images, four maturity classes (B1, B2, B3, and B4), and 4 to 8 side views per tree. The fixed split contains 716 training trees, 96 validation trees, and 141 test trees. Dataset construction, annotation protocol, and per-class composition are documented in [17].

For tree i, the ground-truth target is the count vector

$$\mathbf{y}_i = [\, y_{i,B1},\ y_{i,B2},\ y_{i,B3},\ y_{i,B4} \,], \tag{1}$$

and the prediction is $\hat{\mathbf{y}}_i = [\hat{y}_{i,B1}, \ldots, \hat{y}_{i,B4}]$. The evaluation unit is the tree, not the individual image: per-image appearance counts are not valid BBC outputs.

Let $a_{i,c}$ denote the number of detected appearances of class c across all views of tree i. The naive multi-view estimator

$$\hat{y}_{i,c}^{\text{naive}} = a_{i,c} \tag{2}$$

is biased upward whenever a bunch is visible from more than one side. Under GT annotations, this estimator reaches only 50.00% Class ±1 Acc and 6.38% Tree ±1 Acc on the test split, which motivates learned tree-level aggregation rather than direct summation.

B. Detection Conditions

The same counter is fitted separately under two input conditions: annotated boxes and classes $\rightarrow$ features $\rightarrow$ counts, and cached detector boxes and classes $\rightarrow$ features $\rightarrow$ counts.

In the *GT detection setting*, the feature extractor reads ground-truth bounding boxes and classes directly from the annotation files. Counters trained on these features provide an empirical reference when detections are accurate, rather than a mathematical upper bound on all counters. In the *fixed-detector setting*, the feature extractor reads cached YOLO26m [10] predictions from the released `y26mv2` checkpoint. The detector was trained on the official 716-tree training split for

TABLE I
DETECTOR PERFORMANCE AND APPEARANCE-LEVEL ERRORS.

	Validation		Test split, appearance level				
Class	mAP50	Recall	GT app.	Missed	Wrong	FP	Recall
B1	**0.746**	**0.801**	252	47	33	58	**0.683**
B2	0.425	0.433	496	161	209	158	0.254
B3	0.550	0.656	1409	494	65	393	0.603
B4	0.363	0.389	455	262	89	97	0.229
All	0.521	0.570	2612	964	396	706	0.479

Note: Missed/Wrong use GT classes; FP uses predicted classes. Test All recall is pooled; validation All is macro-averaged.

60 epochs (batch size 32, image size 640, patience 60, seed 42; Ultralytics 8.4.49 in the training log). Inference uses the repository default confidence threshold of 0.25 and the standard Ultralytics post-processing settings. A single detector checkpoint is used for all counter and feature configurations in Sections III-A–III-C, so any observed difference there comes from the counter or feature representation rather than from detector retraining.

Because a claim about detector quality cannot rest on one checkpoint, Section III-E varies the detector along two axes: its operating point, by sweeping the confidence threshold of the same checkpoint, and its capacity, by comparing three earlier YOLO26 checkpoints (n, s, and m) against `y26mv2`. Those three were trained under an earlier 763/95/95 protocol, so they are compared on a checkpoint-fair subset, the 590 trees that train under both protocols and the 64 excluded from gradient training under both. This subset includes validation trees used during detector development; it is not an independent test of a second architecture family. We additionally train YOLO11m on the official 716/96/141 split, using the same epoch, batch, image-size and seed settings, with 12 data-loader workers and Ultralytics 8.4.140. Its validation-selected checkpoint is frozen before test evaluation; Ridge+F_0 is refitted on 716 training trees for each input source. Software versions and architecture-specific defaults differ, so this is a pipeline comparison, not an isolated causal effect of architecture.

Validation mAP50 follows the COCO convention [18].

The test block uses the one-to-one IoU matching in Section III-D: Missed denotes an unmatched annotation, Wrong a matched but misclassified box, and FP an unmatched detection. Recall requires both localisation and class correctness; the validation and test blocks use different protocols.

The per-class weakness on B4 and the moderate recall on B3 in Table I are revisited in Sections III-C and III-D.

C. Feature Vectors

Given either set of detections, observations from all views of a tree are aggregated into a fixed-length feature vector. The 13-dimensional baseline F_0 contains, for each class c, the total count s_c, the per-side maximum m_c, and the per-side mean μ_c, plus the number of available views n_{views}:

$$F_0 = [\, s_{B1:B4},\ m_{B1:B4},\ \mu_{B1:B4},\ n_{\text{views}} \,]. \tag{3}$$

TABLE II
DEFINITIONS OF THE TREE-LEVEL FEATURES.

Group (dim)	Entries	Definition
F_0 (13)	s_c, m_c, μ_c, n_v	total, per-view maximum, and per-view mean detections of class c; n_v is the number of views, 4 or 8 (global)
conf (20)	ΣP_c, $\overline{P}_c$, $\max P_c$, h_c, h_c^+	sum, mean, and maximum confidence; counts with $p \geq 0.5$ and $p \geq 0.6$
spatial (8)	$\overline{cy}_c$, $\overline{A}_c$	mean $cy = (y_{\min} + y_{\max})/(2H)$ and mean $A = wh/(WH)$, for image width W and height H
distrib (20)	σ_c, $\min_v k_{c,v}$, $\sigma_c/(\mu_c+\varepsilon)$, $\lvert\{v\!:\!k_{c,v}\!>\!0\}\rvert$, $1/(1+\sigma_c)$	spread and floor across views, coefficient of variation, views holding a detection, and view-to-view consistency
compos. (6)	T, $s_c/(T+\varepsilon)$, $s_{B3}/(s_{B2}+s_{B3}+\varepsilon)$	all detections $T = \sum_c s_c$ (global), class shares, and the B3 share of the B2/B3 pair (global)

Note: Entries repeat for B1–B4 unless marked global. Empty confidence mean/max and area are 0; empty height is 0.5.

Four optional groups extend F_0: detection confidence (20-dim), spatial position (8-dim), side distribution (20-dim), and cross-class composition (6-dim). Together they form the 67-dimensional combined set

$$F_{\text{all}} = F_0 \cup \text{conf} \cup \text{spatial} \cup \text{distrib} \cup \text{composition}.$$

For n_v views, let $k_{c,v}$ denote the class-c count in view v, P_c its confidence multiset, $s_c = \sum_v k_{c,v}$, $\mu_c = s_c/n_v$, and σ_c the population SD over all views, including zero-count views. All denominators use $\varepsilon = 10^{-6}$. Table II defines every entry so that the representation can be rebuilt from the definition alone. Under GT detections all confidences equal one, so the confidence group degenerates there and the ablation is run only in the fixed-detector setting.

Eight feature sets are evaluated: F_0, F_0+conf, F_0+spatial, F_0+distrib, F_0+conf+spatial, F_0+conf+distrib, F_0+distrib+spatial, and F_{all}.

D. Counter Models and Evaluation Metrics

Two heuristic baselines are included as reference points. The *naive sum* sums all detected appearances per class (defined in Section II-A). The *global divisor* scales the naive sum by a single constant fitted separately in each input condition: the ratio of input appearances to annotated unique bunches over the 716 training trees. This gives $k = 1.891$ for GT inputs and $k = 1.793$ for fixed-detector inputs; neither calibration uses validation or test trees.

Five regression models are evaluated in both detection conditions: Linear Regression, Ridge [19], ElasticNet [20], SVM [21], and Random Forest [22]. Each counter maps a tree feature vector $\mathbf{x}_i$ to the count vector $\hat{\mathbf{y}}_i = f_\theta(\mathbf{x}_i)$.

All fitted counters are trained only on the 716-tree training split with deterministic seed 42, and continuous outputs are rounded to the nearest non-negative integer before scoring. All but Random Forest use standardized features. Ridge uses RidgeCV over $\alpha \in \{0.01, 0.1, 1, 10, 100, 500\}$; ElasticNet uses MultiTaskElasticNetCV with 5-fold cross-validation and 3000 maximum iterations; SVM is an RBF SVR in a multi-output wrapper with 3-fold GridSearchCV over $C \in \{0.1, 1, 10\}$ and $\gamma \in \{\text{scale}, 0.01, 0.1\}$; Random Forest uses 200 trees of maximum depth 10.

These settings preserve the released benchmark. Ridge and ElasticNet scale the outer training data before their internal CV; SVM scales within its CV pipeline. Outer test trees never enter these fits. "Best" below denotes the highest observed test score among compared configurations, not a model selected on an independent validation set.

All counter models are evaluated on the fixed 141-tree test split. Class-level ±1 correctness for tree i and class c is

$$I_{i,c}^{\pm 1} = \begin{cases} 1, & |\hat{y}_{i,c} - y_{i,c}| \le 1, \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

Class $\pm$1 Acc averages this indicator over all test trees and classes:

$$\text{Class } \pm 1 \text{ Acc} = \frac{1}{4N} \sum_{i=1}^{N} \sum_{c=1}^{4} I_{i,c}^{\pm 1}. \tag{5}$$

Tree $\pm$1 Acc is stricter: a tree counts as correct only if all four class predictions are within ±1 simultaneously,

$$\text{Tree } \pm 1 \text{ Acc} = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}\left(\sum_{c=1}^{4} I_{i,c}^{\pm 1} = 4 \right). \tag{6}$$

Macro MAE is the mean absolute error averaged over trees and classes, and **macro RMSE** is the per-class root mean squared error averaged over classes; RMSE is reported because a squared penalty exposes the few badly mis-counted trees that MAE dilutes. **Signed class bias** is the per-class mean $\hat{y}_{i,c} - y_{i,c}$; positive values indicate over-counting, negative values under-counting. Bias is reported alongside accuracy because operational aggregate yield estimates are sensitive to directional error rather than absolute error alone. Two whole-tree aggregate measures complete the set: **total-count MAE**, the mean of $|\sum_c \hat{y}_{i,c} - \sum_c y_{i,c}|$, and **total $\pm$1 Acc**, the fraction of trees whose total bunch count is within one of the truth. These matter for block-level yield forecasting, where per-class assignment can be wrong while the total remains usable.[1]

E. Uncertainty and Validation Protocol

For the main fitted configurations, percentile bootstrap intervals resample trees ($B = 10{,}000$, seed 42); paired intervals reuse the same tree indices. The release includes intervals for per-class and aggregate metrics. Class ±1 differences use an exact paired permutation test exchanging model outcomes within each tree, preserving dependence among its four classes. All ten counter pairs per condition are tested, with Holm adjustment within each condition; the seven Ridge feature comparisons against F_0 form a separate family. Unadjusted p and adjusted p_H are distinguished below. Wilcoxon tests on per-tree macro absolute error and exact McNemar tests on Tree ±1 outcomes address different endpoints and are reported separately. Intervals and tests condition on fitted models and do not account for post-hoc configuration selection or detector retraining. Nonsignificance does not establish equivalence.

A separate selection check ranks all 40 model/feature candidates in each condition on the 96 validation trees: highest Class ±1, then lowest macro MAE, fewer dimensions, and model/feature name as deterministic tie-breaks. Only training trees enter model fitting; test labels do not select the winner. This reanalysis uses the historical test set, not newly collected test data.

Repeated stratified 5-fold CV with 5 repeats uses estate and dominant class jointly for stratification. Matched GT/fixed Ridge+F_0 comparisons use identical folds of the 237 trees outside detector gradient training, fitting on 189–190 trees per fold. Because this pool includes the detector's 96 validation trees, a stricter check uses only the 141 original test trees (112–113 training trees per fold). The detector stays frozen. Fold standard deviations are descriptive, since repeated folds overlap. Cross-estate tests transfer the counter; the detector was trained on both estates.

III. Results and Discussion

Section III-A reports both detection conditions, Section III-B tests whether richer features recover the loss, and Section III-C compares the conditions per class. Section III-D examines detection and aggregation diagnostics, Section III-E tests how far the resulting claim survives changes of detector, split, and site, and Section III-F explains the one counter that behaves differently.

A. Counting Under the Two Detection Conditions

Table III places both conditions side by side. Under GT detections, naive appearance summation yields only 50.00% Class ±1 Acc and 6.38% Tree ±1 Acc, confirming that duplicate visibility cannot be ignored and that direct summation is not a viable estimator. The calibrated divisor recovers most of that loss, and ElasticNet improves further to 98.05%, so with accurate detections the counting problem is simple enough for compact regression models. The four best fitted counters sit within 0.53 percentage points of each other on Class ±1 Acc and 2.13 points on Tree ±1 Acc, and their pairwise Class ±1 differences are not significant ($p_H = 1$). ElasticNet exceeds Random Forest by 2.13 pp (95% CI 0.89–3.55, $p = 0.0042$, $p_H = 0.0376$); SVM also exceeds RF ($p_H = 0.0342$). Section III-F examines RF's prediction behaviour.

For the strongest F_0 Class ±1 scores, the 95% bootstrap intervals are 97.0–99.1% (GT) and 73.2–79.6% (fixed). Moving to cached YOLO26m detections, the best F_0 result (ElasticNet, 76.42%) trails the same counter under GT by 21.63 percentage points (95% CI 18.44 to 25.00, $p < 0.001$). All five counters stay within a 3.19-point window, and none of the ten Class ±1

[1] Source code: https://github.com/ULM-SawitMVC/Baseline-SawitMVC.

TABLE III
COUNTING PERFORMANCE WITH BASELINE FEATURES.

	GT detections			Fixed detector		
Method	Class ±1	Tree ±1	MAE	Class ±1	Tree ±1	MAE
Naive sum	50.00	6.38	2.142	50.71	10.64	2.447
Global divisor	95.57	86.52	0.356	70.21	22.70	1.188
ElasticNet	**98.05**	**92.20**	0.277	**76.42**	29.79	**1.043**
SVM	97.87	91.49	**0.266**	74.82	29.08	**1.043**
Ridge	97.70	90.78	0.275	76.06	28.37	1.053
Linear reg.	97.52	90.07	0.277	75.71	**30.50**	1.048
Random forest	95.92	84.40	0.365	73.23	26.95	1.110

Note: Accuracy is in %; MAE is macro MAE. The divisor is fitted separately per input condition using training trees only.

TABLE IV
RIDGE FEATURE ABLATION UNDER DETECTOR INPUTS.

Features	Dim	Class ±1	Tree ±1	MAE	CV Class ±1
F_0	13	76.06	28.37	1.053	72.70 $\pm$ 2.35
F_0+C	33	75.89	29.79	1.059	**73.26** $\pm$ 2.36
F_0+S	21	76.60	30.50	1.046	72.65 $\pm$ 2.37
F_0+D	33	74.82	28.37	1.060	72.89 $\pm$ 1.96
F_0+C+S	41	76.24	29.08	1.059	72.40 $\pm$ 2.46
F_0+C+D	53	76.42	29.79	1.060	72.59 $\pm$ 2.02
F_0+D+S	41	75.71	30.50	1.048	72.34 $\pm$ 2.33
F_{all}	67	**77.48**	**32.62**	**1.035**	72.34 $\pm$ 2.08

Note: C: confidence; S: spatial; D: distribution. CV: mean $\pm$ fold SD on 237 trees, including detector validation trees. MAE is macro MAE.

comparisons is significant; the widest, ElasticNet over Random Forest, gives $+3.19$ pp (95% CI 0.00 to $+6.38$, $p = 0.0628$, $p_H = 0.628$). A large drop across conditions, set against a spread among counters that the test set cannot resolve, points to the detector output rather than the counter family as the main source of loss.

The per-class picture is the same whichever counter is used: B1 stays near its GT ceiling at 95.74–96.45% Class ±1 Acc, B3 collapses to 55.32–56.03%, and B2 (73.76–80.14%) and B4 (67.38–73.05%) sit at intermediate losses. Detection recall alone does not rank these losses: B3 has the largest counting loss despite higher test recall than B2/B4. Its larger counts and cross-class confusion also affect the absolute ±1 criterion.

B. Feature Ablation

Whether enriched features can recover the detector-driven gap is tested by fixing Ridge and varying the feature configuration (Table IV). On the fixed test split the eight configurations run from 74.82% (F_0+D) to 77.48% (F_{all}) Class ±1 Acc, so the full set gains 1.42 points at class level and 4.25 at tree level over the 13-dimensional baseline. The CV column repeats the eight feature sets using the protocol in Section II-E.

The gain is not statistically resolved. Paired against F_0 it gives $+1.42$ pp (95% CI -0.53 to $+3.37$, $p = 0.201$, $p_H = 1$). Under cross-validation the eight feature sets fall inside a 0.92-point band whose fold-to-fold standard deviation is more than twice as wide, with F_{all} no longer leading (Table IV, last column). No Class ±1 comparison with F_0 is significant after adjustment. This limits the evidence for these particular feature additions; it does not prove that detector outputs contain no further useful information or that all counting methods are equivalent.

TABLE V
PER-CLASS COUNTING PERFORMANCE.

Configuration	Metric	B1	B2	B3	B4	All
GT,	Acc. ±1 (%)	100.00	99.29	93.62	99.29	98.05
ElasticNet$+F_0$	MAE	0.050	0.227	0.603	0.227	0.277
	RMSE	0.223	0.519	0.855	0.491	0.522
	Bias	$-.050$	$+.043$	$-.064$	$-.028$	$-.099$
Fixed,	Acc. ±1 (%)	97.16	82.98	57.45	72.34	77.48
Ridge$+F_{\text{all}}$	MAE	0.369	1.014	1.553	1.206	1.035
	RMSE	0.652	1.437	1.984	1.598	1.418
	Bias	$+.014$	$-.078$	$-.177$	$+.071$	$-.170$

Note: All is the class mean, except Bias, which is total signed error. Counts and errors are in bunches per tree.

C. GT vs. Fixed-Detector Gap

The central comparison is the per-class gap between ideal detections and the fixed detector, reported with the signed bias of the best configuration in each condition in Table V.

The gap is large, at 20.57 percentage points in Class ±1 Acc (95% CI 17.20 to 23.94, $p < 0.001$) and 59.57 points in Tree ±1 Acc, and it is not uniform across maturity classes: B1 loses 2.84 pp, B2 16.31 pp, B3 36.17 pp, and B4 26.95 pp. With the same Ridge$+F_0$ configuration in both conditions, the gap remains 21.63 pp (95% CI 18.44–25.00, $p < 0.001$), so it does not depend on comparing different counter families. The squared-error view agrees: macro RMSE rises from 0.522 to 1.418 and total-count MAE from 0.979 to 1.787 bunches per tree; total ±1 accuracy is 78.01% and 48.94%, respectively. This is per-tree count error; block-level forecasting error also depends on cancellation and dependence across trees.

Validation-based selection chooses Ridge$+F_0$ for GT and SVM$+(F_0 + \text{conf})$ for fixed inputs. Their test Class ±1 accuracies are 97.70% and 74.11%, respectively: a 23.58-point gap (95% CI 20.21–27.13, $p < 0.001$). Their Tree ±1 accuracies are 90.78% and 27.66%. Thus the large between-condition difference persists when test scores do not determine configuration selection.

Bias adds direction to the gap. The fixed-detector Ridge $+F_{\text{all}}$ pipeline under-counts B2 by 0.078 and B3 by 0.177 bunches per tree, while B4 is over-counted by 0.071 despite its low recall. A learned counter combines misses, false positives, and class confusion, so this output bias cannot be assigned to a single confusion direction. The total signed bias is -0.170 bunches per tree; it should be assessed alongside per-tree absolute error before extrapolating to operational harvest forecasts.

D. Detection and Aggregation Diagnostics

Two mechanisms are candidates for the residual: bunches missed in every view and maturity-class confusion. A counter may infer missing counts statistically, but cannot directly associate a bunch with no matched detection. To inspect these mechanisms, every detection on the test trees is matched against the annotations, and each matched detection inherits the identity of the ground-truth bunch it hit. Matching is greedy in order of confidence and class-agnostic, and accepts a pair at IoU ≥ 0.5 in pixel coordinates, so classification errors among matched boxes are separated from unmatched boxes. An unmatched annotation can reflect absence, IoU below threshold, or matching competition; it is not a pure measure of visibility or localisation failure.

The right block of Table I reports the outcome. Of 2,612 annotated appearances on the 141 test trees, 964 (36.9%) are missed outright and 396 (15.2%) are localised but assigned another maturity class, leaving 1,252 (47.9%) both found and labelled correctly; in the other direction, 706 of 2,354 detections (30.0%) match no annotation, at a mean confidence (0.337) well below that of matched detections (0.487). The failure modes are class-specific: B2 loses more appearances to class confusion (209, of which 158 are read as B3) than to missed detection (161), whereas B4 is dominated by outright misses (262 of 455). Counted as physical bunches, 317 of the 1,397 test bunches (22.7%) are never detected in any view, from 8.5% for B1 to 42.3% for B4.

Figure 2(a) compares three diagnostic rules. Counting each matched physical bunch once with its GT class gives 86.17% Class ± 1 Acc. Replacing its GT class by a confidence-weighted vote across matched detections gives 71.81%; adding every unmatched detection as a separate bunch gives 62.23%. Identity is known only for matched detections, so the last rule does not deduplicate false positives across views.

The reductions of 13.83 points from exact GT counts to the first rule, 14.36 between the first two rules, and 9.58 between the last two depend on this ordering, voting rule, and treatment of unmatched detections. They cannot be added to the GT counter's 1.95-point error to explain the deployed counter. Indeed, the deployed 77.48% exceeds two diagnostic rules: learned aggregation can compensate for systematic detector errors. Likewise, $86.17 - 77.48 = 8.69$ points is a difference between estimators, not an upper bound on association or counter improvements; the first rule uses GT classes and removes unmatched detections. The diagnostics expose detection failures, while the matched-input comparison measures their combined effect on a specified counting pipeline. Isolating association error itself would require an implemented association module and controlled interventions on that module.

E. Sensitivity and Resampling

A claim about detector quality drawn from one checkpoint and one split needs additional checks. Table VI compares a second detector family, YOLO26 checkpoint capacities, and confidence thresholds.

On the official 141-tree test set, Ridge+F_0 reaches 73.76% Class ± 1 and 26.24% Tree ± 1 with YOLO11m, versus 76.06% and 28.37% with YOLO26m. The Class ± 1 difference is -2.30 pp (95% CI -4.96 to $+0.35$, exact $p = 0.118$); this does not establish equivalence. GT reaches 97.70%, retaining a 23.94-point gap to YOLO11m (95% CI 20.57–27.30, $p < 0.001$). YOLO11m has 976 unmatched and 399 misclassified appearances out of 2,612 annotations, plus 463 unmatched detections out of 2,099. Thus the GT advantage extends to this additional family, without establishing a general architecture ranking.

The operating-point sweep re-runs inference at confidence 0.05 and filters upward; Ridge+F_{all} is refitted on the 716 training trees at every threshold. Raising the threshold from 0.25 to 0.70 reduces Class ± 1 Acc from 77.48% to 63.12%. At 0.05, 8,807 test detections yield 76.24%. The filtered pass contains 2,353 detections at 0.25 versus 2,354 in the original cache, with the same reported counting accuracies; the two caches are not identical. Threshold 0.10 ties 0.25 on Class ± 1 and reaches 38.30% Tree ± 1 versus 32.62%. This is an exploratory test-set sweep, not independent threshold selection: among these eleven operating points, none exceeds 77.48% Class ± 1, but the whole-tree endpoint can improve.

On the 64 trees outside gradient training under both split protocols, Ridge+F_0 is fitted on the 590 shared training trees. Class ± 1 ranges from 70.70% to 73.83% across four YOLO26 checkpoints, versus 97.27% for GT: a 3.13-point checkpoint spread and a 23.44-point GT gap to `y26mv2`. This within-family comparison confounds capacity with historical training splits and includes detector validation trees; it is not an independent architecture ranking.

Matched 5×5-fold CV with Ridge+F_0 on the same 237-tree pool gives $97.72 \pm 1.02\%$ (GT) versus $72.70 \pm 2.35\%$ (fixed), a mean gap of 25.03 pp. Restricting both conditions to the 141 original test trees gives $97.16 \pm 1.23\%$ versus $74.08 \pm 2.96\%$, a 23.08-point gap. These checks support the ordering with a frozen detector, but their smaller counter training sets prevent interpreting the original gap as conservative. For Ridge+F_{all}, counter transfer from DAMIMAS (641 training trees) to LONSUM (24 evaluation trees) gives 71.88%, versus 72.92% when fitted on LONSUM's 75 training trees. The reverse transfer to 213 DAMIMAS evaluation trees gives 68.78%. These comparisons confound estate and counter training size; the detector saw training trees from both estates.

Vertical-feature sensitivity is tested with the synthetic transformation $\overline{cy} \mapsto \text{clip}(a\overline{cy} + b, 0, 1)$ on test features, with $a \in [0.90, 1.10]$ and $b \in [-0.10, 0.10]$. Missing-class sentinels stay at 0.5, and the counter is trained on unperturbed data. For Ridge+F_{all}, the worst loss is 1.42 pp; additive Gaussian noise with $\sigma = 0.05$ and seed 42 costs 0.53 pp. This checks sensitivity to feature shifts, not physical camera pitch. Actual rotation changes perspective, box area, visibility, and detector outputs; no calibration or reacquired pitched images are available to validate those effects.

F. Why Random Forest Trails

RF's Class ± 1 deficit relative to ElasticNet and SVM is significant under GT detections after Holm correction. A plausible explanation is its prediction range: a forest averages the training targets reaching each leaf, so its output cannot

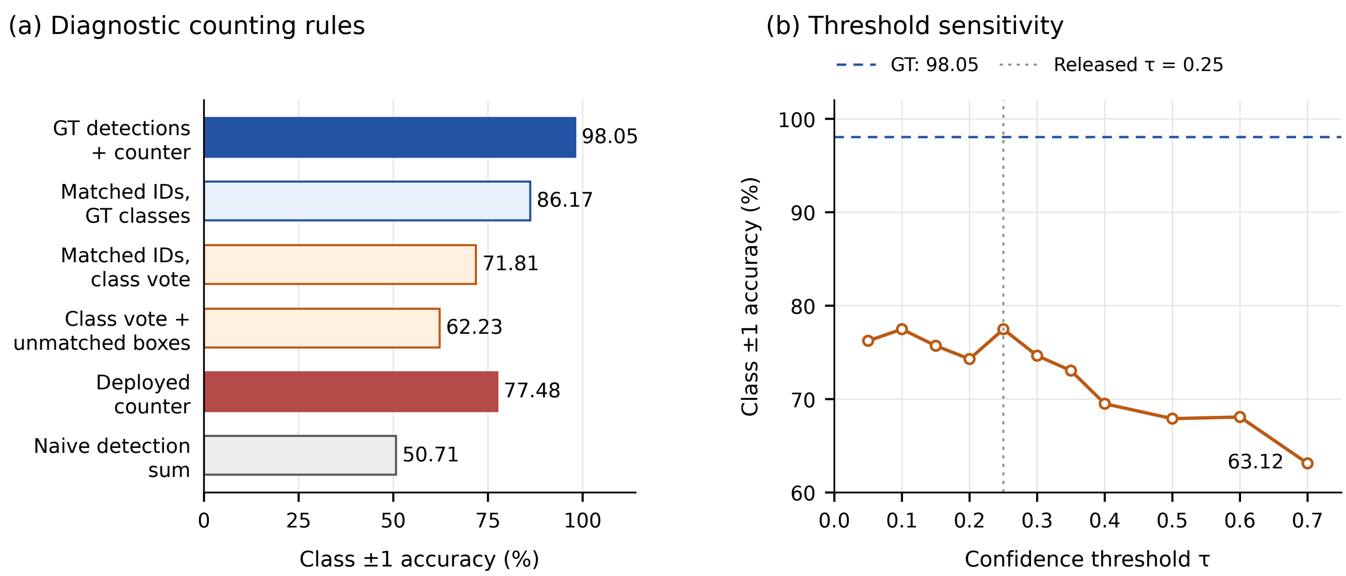

Fig. 2. Detection and aggregation diagnostics: (a) six counting rules; (b) confidence-threshold sensitivity of Ridge+F_{all}. Diagnostic rules are separate estimators, not additive error components.

TABLE VI
DETECTOR FAMILY, CHECKPOINT AND THRESHOLD SENSITIVITY.

Setting	Detections	Recall	Class ±1	Tree ±1
Family, Ridge+F_0, 141 test trees				
YOLO11m	2099	0.445	73.76	26.24
YOLO26m	2354	0.442	76.06	28.37
GT detections	2612	1.000	97.70	90.78
Checkpoint, Ridge+F_0, 64 evaluation trees				
YOLO26n	1013	0.442	72.66	34.38
YOLO26s	825	0.383	72.27	26.56
YOLO26m	950	0.427	70.70	28.12
`y26mv2`	1067	0.442	**73.83**	31.25
GT detections	1203	1.000	97.27	89.06
Operating point, Ridge+F_{all}, 141 test trees				
$\tau = 0.05$	8807	0.591	76.24	34.75
$\tau = 0.10$	5522	0.556	**77.48**	**38.30**
$\tau = 0.25$	2353	0.441	**77.48**	32.62
$\tau = 0.35$	1488	0.355	73.05	23.40
$\tau = 0.50$	799	0.245	67.91	21.28
$\tau = 0.70$	125	0.086	63.12	14.89

Note: Recall is macro class-correct appearance recall at IoU ≥ 0.5; threshold recall is recomputed after filtering. GT counts are annotated appearances; their recall is 1 by construction.

leave the training range and is pulled towards the centre of the target distribution wherever a leaf is thinly populated. Under GT detections the slope of an ordinary least-squares fit of prediction on truth is 0.754–0.888 across the four classes for the forest, against 0.865–0.951 for the linear counters, and the standard deviation of its predictions is a smaller fraction of the true one for every class. The effect is sharpest where counts are small and the mapping is close to affine: for B1 the forest never predicts more than three bunches although the test set holds trees with six, and its MAE is three times that of Ridge (0.149 against 0.050); the B1 training maximum is only five. In the highest-load bin it trails ElasticNet by 3.07 points (94.74% versus 97.81%). These diagnostics are consistent with shrinkage and sparse high-count examples, but do not isolate a cause or exclude improvement from RF tuning. Under the fixed detector, the ElasticNet–RF Class ±1 gap is unresolved ($p = 0.0628$, $p_H = 0.628$).

G. Limitations

Several limitations bound these findings. The benchmark uses one dataset of 953 trees from two estates in South Kalimantan, and the smaller estate contributes 99 trees, so cross-site evidence is suggestive rather than conclusive and absolute numbers may shift under other regions, varieties, or acquisition protocols. The detector evidence covers YOLO26 and one YOLO11m training run; software/default differences and the descriptive 64-tree checkpoint comparison prevent an architecture-only ranking. A plantation excluded from detector training remains untested. Repeated CV refits counters only; it does not estimate detector training variability. Camera-pitch robustness remains unverified: the dataset lacks calibrated pitch measurements or repeated captures at known angles. Synthetic feature shifts cannot reproduce changes in visibility and detector output, so spatial-feature findings are limited to the observed acquisition conditions. Test-based highlighting of configurations limits confirmatory claims; the validation-selection sensitivity check uses the same historical test set. The counters treat each tree independently and ignore plantation-scale spatial or temporal regularities that operational forecasts sometimes exploit. Finally, the B1 through B4 taxonomy admits genuine visual ambiguity at the B2/B3 and B3/B4 boundaries. Without inter-annotator measurements we cannot quantify how much measured confusion arises from label ambiguity. Tree resampling also does not model spatial dependence among neighbouring trees.

IV. Conclusion

This benchmark compares multi-view tree-level FFB counting under GT boxes and cached YOLO26m outputs. The best reported Class ± 1 scores are 98.05% and 77.48%, a 20.57-point gap (95% CI 17.20–23.94). The same Ridge$+F_0$ counter in both conditions retains a 21.63-point gap. No fixed-input counter pair or Ridge feature addition is significant on Class ± 1 after multiplicity correction, although this does not establish equivalence. Matched resampling supports the GT advantage, and identity-based diagnostics reveal substantial missed bunches and class confusion without establishing a causal error budget or a ceiling on counter improvement.

For automated BBC, these findings favour investigating detector recall and maturity confusion alongside aggregation, rather than expecting small feature changes to close the observed gap. Reporting both controlled-input and detector-input results helps distinguish counter behaviour from the input quality it receives. Cached detections, tree splits, and evaluation code make these comparisons reproducible. Per-class and total-count errors further separate maturity allocation from overall inventory error.

The matched-counter GT advantage also holds for YOLO11m (23.94 pp). Evidence remains limited to one dataset, two detector families, and two unequal estates. Future work should improve B3/B4 detection and B2$\to$B3 classification, test explicit cross-view association [6], [7], [23] against learned aggregation under matched inputs, and broaden detector evaluation. An independent plantation, repeated captures at measured camera-pitch angles, and repeated detector training are needed before extending the conclusion to other acquisition conditions or operational harvest forecasts.

Acknowledgment

This research was funded by the Badan Pengelola Dana Perkebunan (BPDP), Indonesia, under contract number PRJ-36/BPDP/2026, through Universitas Lambung Mangkurat (ULM), Banjarmasin, Indonesia.

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